

## Team Project

**Deadline for submission: Last Day of the course**

Late submissions will not be accepted.

### Version 1 - Aragon Municipal Library

| Student Name(s) | Student ID |
|-----------------|------------|
|                 |            |

You will work in a team, with **3-5 students per team**, for the term project database system. Students should form their own teams and send an email to the TA with the names of the team members, if the TA has not assigned you to any teams.

**You can choose MySQL Server or Microsoft SQL Server to develop the project.**

#### **Purpose**

To design and develop a database management system using recommended best practices covered in class. You will work to prepare the final project based on a case problem. You should use all techniques covered in this course.

This project describes a case study project. You must follow all requested tasks and apply them to your own project.

Although one person can maintain a small database management system, larger systems require groups of people filling a variety of roles. Of course, the line between these roles can be blurred, and many aspects of database design require collaboration to solve a problem.

**Project Update Meeting: at the end of each class, you'll have about 15 minutes to discuss your project.**

Start working with your team members. It is often helpful to have someone volunteer to be the "leader". Another group member should volunteer to post the group's SUMMARY message. Some groups divide up the work, and each student has a smaller task. Other groups have each member research and then they combine the best of the answers together. It's up to you.

How to be successful in the group work:

- Determine which group you are in.
- Decide in the next class how to divide the work. Decide who will post the summary response.

#### **Project includes (Part One)**

1. Following the Step-by-Step in Designing a database.
2. Creating database, schema, and table objects (Task 1).
3. Applying data integrity using constraints (Task 2).

#### **Task 1**

To ensure that you have a way to practice everything you learned in Database course, you are going to design and complete a separate database for a similar business, **Aragon Municipal Library**, which has slightly different data needs and requirements from its main business in USA. Aragon Municipal Library is located in Montreal, Quebec (Canada).

This library has similar needs to those of **West Municipal Library, USA**, but because Canada has different requirements for the data it collects, you will need to develop the database with these needs in mind. You will create this database and then do all subsequent work in your copy of the database. This approach makes it possible for you to develop a database from scratch and practice everything you learned in the database courses.

The Library Database Case Study provides an overview of the operations at the **West Municipal Library**, describes the daily library functions, and presents the database that was designed for the library.

Create tables in the Library Database

- I. Create tables in the Library Database, minimum 9 tables for the Library Database:
  1. **Member table**, defining the following column names with their respective data types. *member\_no, lastName, firstName, middleInitial, photograph, and so on.*
  2. **Adult table**, defining the column names with their respective data types. Make sure that the phone\_no column allows null values. Make sure that all other columns do not allow null values. *member\_no, street, city, state, zip, phone\_no, expr\_date, and so on.*
  3. Create the rest of the tables based on the business rules defined in this project. Create the Library diagram.

## Task 2

In this Task, you will:

1. Define and use all types of constraints to enforce business rules: Primary Key, Foreign Key, Default, Check, and Unique constraints to enforce data integrity.
2. Create or modify the default constraint that makes QC (Quebec) the default for the province column in the **adult** table.
3. Create a phone number constraint to the **adult** table. You may use a regular expression pattern.
4. Create all primary key constraints.
5. Write a script that adds all foreign key constraints.
6. Write a script that adds all default constraints.
7. Write a script that adds all check key constraints.
8. Write a script that adds all unique constraints.

## Task 3 – Creating Views

In this task, you will

- I. Use the Library database created and modified in the previous tasks.
  - II. Write scripts that join tables in the library database.
  - III. Use the GROUP BY and HAVING clauses to summarize data by groups.
  - IV. Create Views
1. **Create a mailing list** of Library members that includes the members' full names and complete address information.
  2. **Write and execute a query** on the title, item, and copy tables that returns the **isbn**, **copy\_no**, **on\_loan**, **title**, **translation**, and **cover**, and values for rows in the copy table with an ISBN of 1 (one), 500 (five hundred), or 1000 (thousand). Order the result by ISBN column.
  3. **Write and execute a query** to retrieve the member's **full name** and **member\_no** from the **member** table and the **isbn** and **log\_date** values from the **reservation** table for members 250, 341, 1675. Order the results by **member\_no**. You should show information for these members, even if they have no books or reserve.
  4. **Create a view and save it as adultwideView** that queries the **member** and **adult** tables. Lists the name & address for all adults.
  5. **Create a view and save it as ChildwideView** that queries the **member**, **adult**, and **juvenile** tables. Lists the name & address for the juveniles.
  6. **Create a view and save it as CopywideView** that queries the **copy**, **title** and **item** tables. Lists complete information about each copy.



7. **Create a view and save it as LoanableView** that queries **CopywideView** (3-table join). Lists complete information about each copy marked as loanable (loanable = 'Y').
8. **Create a view and save it as OnshelfView** that queries **CopywideView** (3-table join). Lists complete information about each copy that is not currently on loan (on\_loan = 'N').
9. **Create a view and save it as OnloanView** that queries the **loan**, **title**, and **member** tables. Lists the **member**, **title**, and **loan** information of a copy that is currently on loan.
10. **Create a view and save it as OverdueView** that queries OnloanView (3-table join.) Lists the **member**, **title**, and **loan** information of a copy on loan that is overdue (due\_date < current date).

#### Task 4 Create Stored Procedures and Triggers

11. Generating Usage Reports – seven questions are listed at the end of this document to generate summary information for librarians.

#### Project Delivery and Evaluation

Everything you create should be based on coherent organization, conciseness, and clarity. You must submit all script files used to create the database. There is no need to submit the database files (mdf and ldf).

1. Each team member must submit his copy, which is the same as for other members.
2. Save your database and other documents in a folder with your name and team members.
3. To compress a file/folder, right-click the file/folder, point to Sent To, and then select Compressed (Zipped) Folder.
4. Submit the zipped folder to your instructor via MyCourses services.
5. Grading rubrics are used to determine point totals for the project.

| Website Design Rubric                                                                                                                                                                                           | Percentage |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Crow's Foot or Chen Model</b><br>Logically designed the database design exhibiting a firm understanding of Crow's Foot notation. Completely and accurately explained design usage and purpose                | 5%         |
| <b>Relations and Cardinalities</b><br>Correctly interpreted all the Business Rules and created all necessary relations with clear indication to the cardinalities and observing the three normalizations forms. | 5%         |
| <b>Keys and Entities</b><br>Correctly choose all primary and foreign keys respecting the naming convention                                                                                                      | 5%         |
| <b>Implementation</b><br>Completely populated tables with correct data elements reflecting the ERD design                                                                                                       | 5%         |
| <b>Teamwork</b><br>The team worked well together to achieve objectives. Each member contributed in a valuable way to the project. All data sources indicated a high level of mutual respect and collaboration.  | 5%         |
| <b>Content and Creativity</b><br>The demonstration was imaginative and effective in conveying ideas to the audience.                                                                                            | 5%         |
| <b>Create database using script</b>                                                                                                                                                                             | 5%         |
| <b>Create schemas</b>                                                                                                                                                                                           | 5%         |
| <b>Create tables</b> - Created all of the tables defined in project requirements. Named tables appropriately in regard to their data elements                                                                   | 5%         |
| <b>Apply data integrity to tables with constraints: PK, FK, Unique, Check, Default</b>                                                                                                                          | 5%         |
| <b>Insert Data into Tables</b>                                                                                                                                                                                  | 5%         |
| <b>Manipulate data</b>                                                                                                                                                                                          | 5%         |
| <b>Advanced Join</b>                                                                                                                                                                                            | 5%         |
| <b>Create Clustered and Non-Clustered Indexes</b>                                                                                                                                                               | 5%         |
| <b>Create sub-queries</b>                                                                                                                                                                                       | 5%         |
| <b>Implement views</b>                                                                                                                                                                                          | 5%         |



|                                      |      |
|--------------------------------------|------|
| <b>Create user-defined functions</b> | 5%   |
| <b>Create stored procedures</b>      | 5%   |
| <b>Create triggers</b>               | 5%   |
| <b>Manage transactions</b>           | 5%   |
| <b>Total</b>                         | 100% |

## *Introduction to the Company Database Case Study*

### *Introduction*

The Library Database Case Study provides an overview of the operations at the West Municipal Library, describes the daily library functions, and presents the database that was designed for the library.

### *Overview of Library Operations*

Before a database for librarians and members was implemented, an interview was conducted with the librarians at the West Municipal Library to assess the library's business needs. The decisions that the database made during the design process are explained in the following sections.

### *Daily Library Functions*

Many daily library functions exist. The following are some of the most important.

#### *Uniquely Identifying Books*

Some books may have the same title; therefore, titles cannot be used as a means of identification. Librarians call books **items**. Items are identified by the /international Standard Book Number (ISBN). Books with the same title can have different ISBN numbers if they are in different languages and have different bindings (hard cover or soft cover).

#### *Reserving Books*

If a member wants a book that is out on loan, the book is placed on reserve for them. When the book arrives, a librarian must notify the member who has been waiting the longest. Members can have as many as four books on reserve at one time.

#### *Determining Book Availability*

Librarians must be able to determine how many copies of a book are out on loan at any given time and which books are on reserve.

A synopsis that ranges from one sentence to several pages exists for each title in the library. Librarians want to be able to access the synopses when members request information about books.

#### *Enrolling Members*

To become a library member, individuals must provide their mailing addresses and phone numbers. A librarian then issues the individual a numbered, machine-readable card. This card is good for one year.

Juveniles (individuals under age 18) can be members of the library, but an adult member must sign for them when they join. Therefore, a juvenile member's card is good only until the associated adult member's card expires. The only information that the library keeps on juvenile members is their name and date of birth. The library must be able to detect when juvenile members turn 18 and then must automatically convert the juvenile memberships to adult memberships.

A librarian must notify the member, a month before membership cards expire,

#### *Checking Out Books*

Books can be checked out for 14 days. Members are allowed to have only four books checked out at a time. If a book is overdue, members have one week before the library sends a notice to them.



Members bring books to the front desk after they locate the ones that they want to check out. A librarian then runs the members card through a machine that reads the card number magnetically. A screen displays information about the member's account, such as name, address, phone number, and the card's expiration date. Ideally, cards that have expired or are about to expire will be highlighted.

The screen also displays information about a member's outstanding loans, including title, checkout date, and due date. This information is useful because it is presented in a chronological sequence, with the most overdue loan appearing first and the most recent loan appearing last. Highlighting also indicates loans that are overdue or are about to become overdue.

If a member's account is in order, a librarian checks out the books. Librarians check out books by running a scanner down the book spines (The ISBN and the copy number are encoded on the spines). The ISBN, title, and the information then appear on the computer screen. If the books are not loanable, a warning message appears.

### ***Checking In Books***

When books are returned, librarians check them in by running a scanner down the book spines. The ISBN number, title, and author information then appear on the computer screen, as well as the member number and name and the book's due date.

Occasionally, books are accidentally re-shelved before librarian check them in. If a member tries to check out a book that the database lists as checked out, librarians need to be able to access the checkout information, including the member's name, check out date, and due date. If a member presents a book to check out that is still officially checked out to another member, a message appears that alerts librarians that the book is already checked out. Then librarians can update their records immediately by being forced to clear the previous loan before they continue with the checkout.

### ***Generating Usage Reports***

Occasionally, librarians must compile usage information, mostly for the Town Council or the Planning Commission. These groups usually want to know information, such as the volume of circulation, the popularity of various books, the reliability of return, and the average length of a borrowing term. Therefore, the librarians need to be able to prepare quick summaries of this information.

The types of questions that are frequently asked include the following:

1. How many loans did the library make last year?
2. What percentage of the membership borrowed at least one book?
3. What was the greatest number of books borrowed by any one individual?
4. What percentage of the books was loaned out at least once last year?
5. What percentage of all loans eventually become overdue?
6. What is the average length of a loan?
7. What are the library's peak hours for loans?

The answers could be created as database objects: views, functions, stored procedures, or triggers it's up to you to choose what to design for each question, but you must cover all objects.

